



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 1804
CALIBRATION DATE: 24-Feb-26

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

a0 = 1.799298e-05
a1 = 2.701622e-04
a2 = -2.049104e-06
a3 = 1.490291e-07

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	703386.5	1.0000	0.0000
4.5000	602000.2	4.5000	-0.0000
15.0000	384702.0	14.9999	-0.0001
18.5000	333378.8	18.5001	0.0001
24.0000	267763.7	24.0002	0.0002
29.0000	220704.5	28.9997	-0.0003
32.5000	193396.5	32.5002	0.0002

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature

